

DSAT FOUNDATION COURSE

Math Theory & Practice

Building the Foundation for the PSAT — and Beyond

- 29 skills across Algebra, Problem-Solving & Data Analysis, Advanced Math, and Geometry & Trigonometry
- Clear, bullet-point theory for every skill — no dense paragraphs
- One representative Medium-tier worked problem per skill, from the companion's 4 Easy + 4 Medium practice set
- Greek terminology on every slide — precise vocabulary is the first step to understanding a question

The 29 Skills

1. Lines
2. Systems of Linear Equations
3. Linear Inequalities
4. Absolute Value
5. Exponent Rules
6. Radicals, Rational Exponents & Scientific Notation
7. Ratios & Proportions
8. Probability
9. Unit Conversions & Rates
10. Statistics — Center & Spread
11. Data Displays — Histograms, Dot Plots & Box Plots
12. Scatterplots
13. Sampling, Bias & Margin of Error

The 29 Skills (continued)

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|---|--|----|--|
| 1 | 14. Functions — Notation, Domain, Range & Graphs | 9 | 22. Radicals & Rational Exponents |
| 2 | 15. Composite & Inverse Functions | 10 | 23. Exponential Functions — Growth & Decay |
| 3 | 16. Function Transformations | 11 | 24. Radical & Rational Equations |
| 4 | 17. Polynomial Operations & Special Products | 12 | 25. Angles, Parallel Lines & Triangle Fundamentals |
| 5 | 18. Factoring Polynomials | 13 | 26. Right Triangles & Trigonometry |
| 6 | 19. Dividing Polynomials & Rational Expressions | 14 | 27. Radians & the Unit Circle |
| 7 | 20. Solving Quadratic Equations & the Discriminant | 15 | 28. Circles — Circumference, Arcs, Sectors, Equations & Theorems |
| 8 | 21. Quadratic Graphs, Roots & Word Problems | 16 | 29. Volume, Surface Area & Similarity of Solids |



BEFORE WE BEGIN

How to Use the Desmos Calculator

The Digital SAT gives you a built-in Desmos calculator for the whole Math section, inside Bluebook. Learn these six moves now — every Desmos Tip in this course assumes you know them.



Typing, Graphing & Tables

- Typing & Calculating: click into a line and type any expression — Desmos calculates it instantly, like a scientific calculator.
- Typing & Calculating: use ^ for exponents, $\sqrt{\quad}$ for square roots, and / for fractions — Desmos formats them as you type.
- Graphing: type an equation with x and y (e.g. $y = 2x + 3$) and Desmos plots it immediately on the grid.
- Graphing: drag to pan and pinch/scroll to zoom, then click a point on the graph to read its exact coordinates.
- Tables: click + and choose 'table,' or type x1, y1 — link it to a typed function to fill in matching y-values.
- Tables: a table is the fastest way to test a few values without doing the arithmetic by hand.

Golden Rule: Practice on the Digital SAT's actual Bluebook Desmos calculator, not a lookalike app — the interface is slightly different everywhere else.

Intersections, Sliders & Regression

- Intersections: graph both equations on the same screen, then click the point where the two graphs cross.
- Intersections: Desmos labels the exact coordinates — that point is the solution to the system.
- Sliders: type an equation with an undefined constant (e.g. $y = ax + 1$) and Desmos offers to make 'a' a slider.
- Sliders: drag the slider to watch how changing that value moves or reshapes the graph in real time.
- Regression: enter data points as a table, then type $y_1 \sim mx_1 + b$ beneath it to fit the line of best fit.
- Regression: use this on scatterplot questions to get the equation of the best-fit line directly, not by eye.

Golden Rule: Reach for Desmos when graphing or tabling is faster than solving by hand — then always re-read the question before you answer.



PART 1

Algebra

Lines • Systems of Linear Equations • Linear Inequalities • Absolute Value • Exponent Rules • Radicals, Rational Exponents & Scientific Notation

What It Is & Core Rules

- A line is any relationship where y changes by the same fixed amount every time x increases by 1.
- That fixed rate of change is called the slope, and it's the single most important number for any line.
- The SAT tests lines because they model steady, predictable change, like a constant hourly rate or a fixed fee.
- Almost every Lines question is really just asking you to find or use slope, an intercept, or both.
- You'll see lines as equations, as tables of points, as graphs, and as word problems — same math, different disguise.
- Slope Formula — $m = (y_2 - y_1) / (x_2 - x_1)$
- Reading Slope's Sign and Special Cases
- Slope-Intercept Form — $y = mx + b$
- Point-Slope Form — $y - y_1 = m(x - x_1)$
- Standard Form — $Ax + By = C$
- Parallel and Perpendicular Lines — Parallel: $m_1 = m_2$. Perpendicular: $m_1 \times m_2 = -1$
- Finding Intercepts — x-intercept: set $y = 0$. y-intercept: set $x = 0$.

Common Mistakes: Subtracting coordinates in mismatched order ($y_1 - y_2$ over $x_2 - x_1$) flips the sign of the slope — always match the order. 
Reading slope by counting grid squares on a graph instead of using real coordinate values — axes can be scaled unevenly.

A vintage record shop in Berlin tracks its weekly profit. In week 3, its profit was -120 euros (a loss). In week 9, its profit was 240 euros. Assuming profit changes at a constant rate, in which week does the shop first break even (profit = 0)?

Formula / Rule Used: Slope formula + point-slope form, then set $y = 0$.

Work through it, then advance for the full step-by-step solution.

Step 1: Treat the two situations as points: (3, -120) and (9, 240), where x is the week and y is profit.

Step 2: Find the slope: $m = (240 - (-120)) / (9 - 3)$.

Step 3: Simplify the top: $240 + 120 = 360$, so $m = 360 / 6$.

Step 4: Divide: $m = 60$ (profit grows by 60 euros each week).

Step 5: Use point-slope form with (3, -120): $y - (-120) = 60(x - 3)$.

Step 6: Simplify: $y + 120 = 60x - 180$.

Step 7: Solve for y: $y = 60x - 300$.

Step 8: Set $y = 0$ to find break-even: $60x = 300$, so $x = 5$.

Answer: Week 5

Tip: Whenever two data points are given and the question asks 'when does it hit zero,' build the full linear equation first.

Trap: It's tempting to just average the two given weeks (3 and 9) instead of actually solving for where profit equals zero.

Tempting Wrong Answers — and Exactly What Went Wrong

Week 6

Averaged the two given weeks, $(3 + 9)/2 = 6$, instead of solving the equation for profit = 0.

Week 1

Made a sign error simplifying $y + 120 = 60x - 180$, writing $y = 60x - 60$ instead of $y = 60x - 300$.

Week 9

Mistook the second data point, where profit is already positive, for the break-even week.

Shortcut: The direct method is fastest here — building the linear equation and setting it to zero avoids error-prone repeated addition.

Desmos Tip: Type $y = 60x - 300$ in Desmos and trace the graph to find where it crosses the x-axis, confirming week 5.

What It Is & Core Rules

- A system of linear equations is just two straight-line equations considered together.
- Solving the system means finding the (x, y) point that makes BOTH equations true at once.
- On a graph, that point is exactly where the two lines cross.
- The SAT tests this two ways: actually solving for x and y, and reasoning about how many crossing points exist.
- You'll see this skill in both bare algebra questions and real-world 'two unknowns' word problems.
- Substitution Method — Solve one equation for a variable, plug it into the other
- Elimination Method — Add or subtract the equations to cancel a variable
- Setting Equal (when both solved for y) — If $y = \dots$ and $y = \dots$, set the right sides equal
- Solving Directly for a Combined Expression — Add, subtract, or scale the equations to build the target
- Number of Solutions — One: different slopes. None: same slope, different y-intercept. Infinite: same line.
- Coefficient-Ratio Shortcut — No solution: $A1/A2 = B1/B2 \neq C1/C2$. Infinite: $A1/A2 = B1/B2 = C1/C2$.

Common Mistakes: Solving for only one variable and forgetting to finish by finding the second one. ⚠ Substituting an expression into the SAME equation it came from instead of the other equation.

The system of equations $6x + 9y = 15$ and $4x + ky = 8$ has no solution. What is the value of k ?

Formula / Rule Used: Coefficient-ratio shortcut: for no solution, $A1/A2 = B1/B2$, but not equal to $C1/C2$.

Work through it, then advance for the full step-by-step solution.



Step 1: A system has no solution when the two equations represent parallel lines — same slope, different y-intercept.

Step 2: In standard form $Ax + By = C$, this means the ratio of the A's equals the ratio of the B's, but NOT the ratio of the C's.

Step 3: Set up the ratio of the x-coefficients: $6/4$, which simplifies to $3/2$.

Step 4: For no solution, the y-coefficients must share this same ratio: $9/k = 3/2$.

Step 5: Cross-multiply: $9 \times 2 = 3 \times k$.

Step 6: Simplify the left side: $18 = 3k$.

Step 7: Divide both sides by 3: $k = 6$.

Step 8: Quick check: with $k = 6$, both equations simplify to $2x + 3y = 5$ and $2x + 3y = 4$ — same left side, different right side, confirming no solution.

Answer: $k = 6$

Tip: For a 'no solution' or 'infinite solutions' question with an unknown constant, set up the coefficient ratio instead of solving the whole system.

Trap: It's tempting to set the y-coefficient ratio equal to the constant ratio ($9/k = 15/8$) instead of the x-coefficient ratio — that's the infinite-solutions setup, not the no-solution one.

Tempting Wrong Answers — and Exactly What Went Wrong

k = 4.8

Used the constant ratio $15/8$ instead of the x-coefficient ratio $6/4$ for the proportion, mixing up the no-solution and infinite-solutions conditions.

k = 13.5

Set up the proportion upside down, writing $k/9 = 6/4$ instead of $9/k = 6/4$.

k = 1.5

Mistook the simplified ratio $3/2$ itself for the value of k, without actually solving the proportion $9/k = 3/2$.

Shortcut: Skip solving the full system entirely — setting the coefficient ratios equal (or unequal) as a proportion is always faster for these constant-finding questions.

Desmos Tip: Type the first equation as $6x+9y=15$ and the second as $4x+6y=8$ (using $k=6$) — Desmos draws two parallel, non-overlapping lines, confirming no intersection.

What It Is & Core Rules

- An inequality compares two expressions using $<$, $>$, $\leq$, or $\geq$ instead of an equals sign.
- It describes a whole range of possible values, not just one single exact answer.
- The SAT tests inequalities because many real situations have limits, minimums, or maximums, not exact targets.
- You solve an inequality almost exactly like an equation, with one key exception to watch for.
- The answer is usually a range like $x > 3$, or a shaded region on a graph.
- Solving Like an Equation — Add or subtract the same value on both sides; the sign stays the same.
- The Flip Rule — Multiply or divide both sides by a negative number $\rightarrow$ flip the sign.
- Compound (Three-Part) Inequalities — $a < x < b$
- Graphing on a Number Line
- Graphing Inequalities in the xy -Plane — $y > mx + b$, $y < mx + b$, $y \geq mx + b$, $y \leq mx + b$
- Systems of Inequalities
- Translating Word Problems — 'at least' $\rightarrow \geq$, 'at most' $\rightarrow \leq$, 'more than' $\rightarrow >$, 'fewer/less than' $\rightarrow <$

Common Mistakes: Forgetting to flip the sign when multiplying or dividing both sides by a negative number. ⚠ Flipping the sign when only adding or subtracting, which is never actually required.

The inequality $2x + k \leq 10$ has the solution set $x \leq 3$. What is the value of k ?

Formula / Rule Used: The boundary of the solution set makes the inequality true as an equation.

Work through it, then advance for the full step-by-step solution.



Step 1: Since the solution set is $x \leq 3$, the boundary value $x = 3$ must make the inequality an equation: $2(3) + k = 10$.

Step 2: Multiply: $6 + k = 10$.

Step 3: Subtract 6 from both sides: $k = 10 - 6$.

Step 4: Simplify: $k = 4$.

Step 5: Check the answer: $2x + 4 \leq 10$ leads to $2x \leq 6$, which gives $x \leq 3$, matching the given solution set.

Answer: 4

Tip: Whenever you're given the solution set instead of the constant, plug the boundary value in as if it were an equation.

Trap: Don't try to guess-and-check random values for k — setting the boundary value equal is faster and guaranteed to work.

Tempting Wrong Answers — and Exactly What Went Wrong

-4

Set up $6 + k = 10$ but made a sign error, writing $k = 6 - 10$ instead of $k = 10 - 6$.

10

Plugged $x = 3$ in but forgot to multiply 2 times 3, treating the equation as $x + k = 10$.

16

Added 6 and 10 together instead of subtracting, treating the equation as $k = 6 + 10$.

Shortcut: Since you're given the exact boundary value, plugging it in as an equation is already the fastest possible method.

Desmos Tip: Type $y1 = 2x + 4$ and $y2 = 10$ in Desmos to confirm the first line stays at or below the second exactly for $x \leq 3$.

What It Is & Core Rules

- Absolute value means 'distance from zero' on the number line.
- Distance is never negative, so $|x|$ is always zero or positive.
- The bars $| |$ just strip away any negative sign once distance is measured.
- The SAT tests this because two different numbers can have the same absolute value.
- That 'two numbers, one distance' idea is why absolute value problems often have two answers.
- Solving $|X| = a$ — If $a \geq 0$: $X = a$ or $X = -a$. If $a < 0$: no solution.
- Absolute Value Inequalities (less than) — $|X| < a \rightarrow -a < X < a$
- Absolute Value Inequalities (greater than) — $|X| > a \rightarrow X < -a$ or $X > a$
- $|A| = |B|$ — $A = B$ or $A = -B$
- Checking for Extraneous Solutions
- Distance Interpretation — $|x - a|$ = distance between x and a
- Absolute Value as a Function — $f(x) = |x|$ has a V-shaped graph with its vertex at the origin.

Common Mistakes: Forgetting the negative case and writing only one solution instead of two. ⚠ Flipping $<$ and $>$ rules — mixing up when it's a 'between' range vs. a 'split' range.

The solution to the inequality $|x - h| < k$ is the range $5 < x < 17$. What is the value of $h \times k$?

Formula / Rule Used: Reverse of $|X - h| < k$: the range is $h - k < X < h + k$, so h is the midpoint and k is the half-width.

Work through it, then advance for the full step-by-step solution.



Step 1: Compare the given compound inequality $5 < x < 17$ to the general form $h - k < x < h + k$.

Step 2: The midpoint of the range gives h , so add the two endpoints: $5 + 17 = 22$.

Step 3: Divide by 2 to find h : $h = 11$.

Step 4: The distance from the midpoint to either endpoint gives k , so subtract: $k = 17 - 11$.

Step 5: Simplify: $k = 6$.

Step 6: Check with the lower endpoint: $h - k = 11 - 6 = 5$, which matches, confirming both values.

Step 7: Multiply to answer the question asked: $h \times k = 11 \times 6 = 66$.

Answer: $h \times k = 66$

Tip: When you're given the solved range, find the midpoint first — that's always h — then measure the distance to either end to get k .

Trap: It's tempting to grab 17 or 5 directly as h without averaging the endpoints — always find the true midpoint first.

Tempting Wrong Answers — and Exactly What Went Wrong

60

Mistook the lower bound 5 for h instead of averaging 5 and 17 to find the true midpoint 11.

132

Used the correct midpoint $h = 11$ but used the full range width $17 - 5 = 12$ as k instead of half the width.

204

Used the upper bound 17 as h and the full range width 12 as k , multiplying two wrong values together.

Shortcut: If only h were needed, averaging the two endpoints would be enough; since this asks for h times k , both values must be found individually first.

Desmos Tip: Type $y = \text{abs}(x - 11)$ and $y = 6$ in Desmos; the graph crosses $y = 6$ exactly at $x = 5$ and $x = 17$, confirming $h = 11$ and $k = 6$.

What It Is & Core Rules

- An exponent is just shorthand for repeated multiplication of the same number.
- a^n means 'multiply a by itself n times,' where a is the base and n is the exponent.
- The SAT tests whether you can simplify exponent expressions using a small set of fixed rules.
- It also tests solving equations where the variable sits up in the exponent itself.
- Every exponent rule on the test can be rebuilt from scratch just by expanding the multiplication.
- Product Rule (multiplying same base) — $a^m \cdot a^n = a^{m+n}$
- Quotient Rule (dividing same base) — $a^m / a^n = a^{m-n}$
- Power Rules (a power raised to a power) — $(a^m)^n = a^{m \cdot n}$
- Power Rules (product or quotient raised to a power) — $(ab)^n = a^n \cdot b^n$ $(a/b)^n = a^n / b^n$
- Zero Exponent — $a^0 = 1$ (as long as $a \neq 0$)
- Negative Exponents — $a^{-n} = 1/a^n$
- Solving Exponential Equations (matching bases) — If $a^m = a^n$ and a is the same on both sides, then $m = n$.

Common Mistakes: Adding exponents when the bases are different — the product rule only works for matching bases. ⚠ Treating a negative exponent as a negative number instead of a reciprocal — fix: flip it to $1/a^n$.



If a is a positive number and $a^5 = 32$, what is the value of a^{10} ?

Formula / Rule Used: Power of a power rule, used in reverse.

Work through it, then advance for the full step-by-step solution.

Step 1: Notice that $a^{(10)}$ can be rewritten as $(a^{(5)})^{(2)}$, since multiplying the exponents 5×2 gives 10.

Step 2: You're already given that $a^{(5)} = 32$, so substitute that value in directly.

Step 3: This gives $a^{(10)} = (a^{(5)})^{(2)} = 32^{(2)}$.

Step 4: Compute $32^{(2)}$: $32 \times 32 = 1024$.

Step 5: So $a^{(10)} = 1024$, without ever needing to find a itself.

Answer: 1024

Tip: Before solving for the base itself, check whether you can rewrite the target expression using the given power directly.

Trap: Many students solve $a^{(5)} = 32$ for a first ($a = 2$), then compute $2^{(10)}$ — that works but takes longer and risks a slip.

Tempting Wrong Answers — and Exactly What Went Wrong

64

Treated $a^{(10)}$ as $2 \cdot a^{(5)}$ (doubling) instead of squaring $a^{(5)}$, giving $2 \times 32 = 64$.

20

Solved for $a = 2$ correctly but then multiplied 2×10 instead of computing $2^{(10)}$.

320

Multiplied 32 by 10 instead of recognizing that $a^{(10)} = (a^{(5)})^{(2)}$, not 10 times $a^{(5)}$.

Shortcut: Recognizing $a^{(10)} = (a^{(5)})^{(2)}$ and substituting 32 directly is the shortcut — it skips solving for a entirely.

Desmos Tip: Type $2^{(5)}$ into Desmos to confirm it equals 32 (so $a=2$), then type $2^{(10)}$ to confirm the answer 1024.

What It Is & Core Rules

- A radical, like a square root or cube root, asks 'what number times itself gives me this?'
- A rational exponent, like $x^{1/2}$, is just another way to write that exact same root.
- Scientific notation is a compact way to write very large or very small numbers using powers of 10.
- The SAT tests these together because they're all really the same exponent rules in different disguises.
- You'll need to convert smoothly between radical form, exponent form, and scientific notation on the same question.
- Radical-Exponent Conversion — n -th root of $(a^m) = a^{m/n}$
- Simplifying Radicals — $\sqrt{xy} = \sqrt{x} \cdot \sqrt{y}$
- Exponent Rules Recap — $a^m \cdot a^n = a^{m+n}$ | $a^m / a^n = a^{m-n}$ | $(a^m)^n = a^{mn}$
- Negative and Zero Exponents — $a^{-n} = 1/a^n$ and $a^0 = 1$
- Combining Radicals with Different Indexes
- Solving Equations with Rational Exponents — If $x^{m/n} = k$, then $x = k^{n/m}$
- Scientific Notation — $N \cdot 10^n$, where $1 \leq N < 10$

Common Mistakes: Treating $\sqrt{x+y}$ as $\sqrt{x} + \sqrt{y}$ — a root never distributes over addition or subtraction. ⚠ Reading a negative exponent as making the value negative instead of taking its reciprocal.



If $x^{(2/3)} = 8^{(4/3)}$, what is the value of x ?

Formula / Rule Used: Simplify one side to a number, then raise both sides to the reciprocal exponent.

Work through it, then advance for the full step-by-step solution.



Step 1: First simplify the right side: $8^{(\frac{4}{3})}$ means (cube root of 8) raised to the 4th power.

Step 2: The cube root of 8 is 2, since $2^3 = 8$.

Step 3: Raise that result to the 4th power: $2^4 = 16$.

Step 4: The equation now reads $x^{(\frac{2}{3})} = 16$.

Step 5: To isolate x, raise both sides to the reciprocal exponent, $\frac{3}{2}$.

Step 6: $x = 16^{(\frac{3}{2})}$ means (square root of 16) raised to the 3rd power.

Step 7: The square root of 16 is 4, so $4^3 = 64$.

Step 8: So $x = 64$.

Answer: $x = 64$

Tip: Simplify the known side down to a single plain number first, then flip the exponent to solve for the variable.

Trap: Using the same exponent $\frac{2}{3}$ to solve for x instead of correctly flipping it to its reciprocal, $\frac{3}{2}$, undoes nothing.



Tempting Wrong Answers — and Exactly What Went Wrong

$x = 16$

Correctly simplified the right side to 16 but stopped there, forgetting to actually solve $x^{(2/3)} = 16$ for x .

$x = 12$

Found the square root of 16 correctly as 4, but then multiplied by 3 instead of raising it to the 3rd power.

$x = 32/3$

Treated the exponent $4/3$ as a multiplier on 8 instead of a root-and-power operation, and stopped there.

Shortcut: Rewriting 8 and 16 as powers of 2 ($8 = 2^3$), $16 = 2^4$) lets you compare exponents directly instead of computing roots.

Desmos Tip: Graph $y = x^{(2/3)}$ and $y = 16$ in Desmos and find their intersection point — the x-coordinate confirms $x = 64$.



PART 2

Problem-Solving and Data Analysis

Ratios & Proportions • Probability • Unit Conversions & Rates • Statistics • Data Displays • Scatterplots • Sampling, Bias & Margin of Error

What It Is & Core Rules

- A ratio compares two quantities by division, like comparing boys to girls in a class.
- A proportion is simply two ratios set equal to each other, like $a/b = c/d$.
- The SAT tests ratios because real life is full of scaled comparisons: recipes, maps, mixtures, speeds.
- A ratio only describes a relationship between parts, not the actual sizes of those parts.
- Once you know the relationship, you can scale it up or down to find any real quantity.
- Ratio Notation — $a : b = a/b$
- The 'x' Method (Common Multiplier) — If $a:b = m:n$, then $a = mx$ and $b = nx$ for some number x .
- Proportions & Cross-Multiplication — $a/b = c/d \iff ad = bc$
- Combining Two Ratios (Ratio Chains)
- Part-to-Part vs. Part-to-Whole
- Direct vs. Inverse Proportion — Direct: $y = kx$. Inverse: $y = k/x$.
- Scaling a Ratio (Equivalent Ratios) — $a:b = ka:kb$ for any k

Common Mistakes: Mixing up the order of a ratio, writing $b:a$ when the question actually asked for $a:b$. ⚠ Treating a part-to-part ratio as if it were already a fraction of the whole.

At a summer camp in Crete, the ratio of counselors to junior counselors is 2:5, and the ratio of junior counselors to campers is 3:8. If there are 240 campers at the camp, how many counselors are there?

Formula / Rule Used: Combine two ratios into a chain using their shared term, then apply the 'x' method.

Work through it, then advance for the full step-by-step solution.

Step 1: The shared term between the two ratios is junior counselors: 5 in the first ratio, 3 in the second.

Step 2: Find the LCM of 5 and 3, which is 15, so scale each ratio to make its junior counselors term equal 15.

Step 3: Scale the first ratio (2:5) by 3: counselors : junior = 6 : 15.

Step 4: Scale the second ratio (3:8) by 5: junior : campers = 15 : 40.

Step 5: Combine both into a single chain: counselors : junior : campers = 6 : 15 : 40.

Step 6: Let campers = $40x$, so $40x = 240$, giving $x = 6$.

Step 7: Find the counselors: $6x = 6(6) = 36$.

Answer: 36 counselors

Tip: When two ratios share a common category, scale both so that shared number matches before combining them into one chain.

Trap: It's tempting to compare counselors to campers directly (2 to 8) without matching the junior counselors term first, which gives a wrong ratio.

Tempting Wrong Answers — and Exactly What Went Wrong

60 Compared counselors to campers directly as 2:8 without scaling through the shared junior counselors term, computing $240 \times 2/8$.

90 Combined the ratios correctly but reported the junior counselors count ($15x = 90$) instead of the counselors count.

30 Found $x = 6$ correctly but multiplied by 5 instead of 6, using the un-scaled ratio number by mistake.

Shortcut: None faster here — combining the ratios into one chain before using the total is necessary since counselors and campers aren't directly compared.

Desmos Tip: Type $y1 = 6x$, $y2 = 15x$, and $y3 = 40x$ in Desmos, then find x where $y3 = 240$ to instantly read off $y1 = 36$.

What It Is & Core Rules

- Probability is a number between 0 and 1 that measures how likely an event is to happen.
- 0 means the event is impossible, 1 means it's certain, and everything else is somewhere in between.
- The SAT tests probability because it shows up in real decisions, like odds, surveys, and games.
- Most SAT probability questions give you counts (how many of each thing) rather than abstract formulas.
- You'll see single events, combined events with 'and'/'or', and probability read straight off a data table.
- Basic Probability — $P(\text{event}) = \text{favorable outcomes} / \text{total outcomes}$
- Probability Is Always Between 0 and 1 — $0 \leq P(\text{event}) \leq 1$
- The Complement Rule — $P(\text{not } A) = 1 - P(A)$
- 'And' — Two Events Both Happening — Independent: $P(A \text{ and } B) = P(A) \times P(B)$
- 'And' With Dependent Events — Dependent: $P(A \text{ and } B) = P(A) \times P(B \text{ after } A)$
- 'Or' — At Least One of Two Events — $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- Probability From a Two-Way Table — $P = (\text{cell value}) / (\text{row total, column total, or grand total})$

Common Mistakes: Using the original total instead of the reduced total for a 'without replacement' second draw. ⚠ Forgetting to subtract the overlap in an 'or' problem, which double-counts outcomes and inflates the answer.

A box of USB drives at an electronics store in Seoul contains only red and green drives, 24 drives in total. The probability of randomly selecting a red drive from the box is $\frac{3}{8}$. If two drives are selected at random without replacement, what is the probability that both selected drives are green?

Formula / Rule Used: Dependent events (without replacement): $P(A \text{ and } B) = P(A) \times P(B \text{ after } A)$.

Work through it, then advance for the full step-by-step solution.

Step 1: Find the number of red drives: $(3/8) \times 24 = 9$ red drives.

Step 2: Subtract from the total to find the green drives: $24 - 9 = 15$ green drives.

Step 3: Find the probability the first drive selected is green: $15/24$.

Step 4: Since the first drive is not replaced, one green drive and one total drive are now gone.

Step 5: The green drives remaining is $15 - 1 = 14$, and the total remaining is $24 - 1 = 23$.

Step 6: Find the probability the second drive is also green: $14/23$.

Step 7: Multiply the two dependent probabilities: $(15/24) \times (14/23)$.

Step 8: Simplify $15/24$ to $5/8$ first, then multiply: $(5/8) \times (14/23) = 70/184 = 35/92$.

Answer: 35/92

Tip: When a probability and a total are given, first find the actual counts of each group before doing anything else.

Trap: It's tempting to treat this like 'with replacement' and use 24 as the total for both draws, instead of reducing it for the second draw.

Tempting Wrong Answers — and Exactly What Went Wrong

25/64

Treated the two draws as independent (with replacement), computing $(\frac{15}{24})^2 = \frac{25}{64}$ instead of reducing the total after the first draw.

3/23

Used the red-drive count (9) as if it were the green-drive count, computing $(9/24) \times (8/23)$ instead of using 15 green drives.

35/96

Correctly reduced the number of green drives left (14) but forgot to also reduce the total drives left, using 24 again instead of 23.

Shortcut: None beyond finding the green count first and multiplying the two dependent probabilities directly — that's already the fastest path.

Desmos Tip: Type $(15/24) * (14/23)$ into the Desmos calculator to confirm the decimal matches the simplified fraction $35/92$.

What It Is & Core Rules

- Unit conversion means rewriting the same real quantity using a different unit, without changing its actual size.
- The SAT tests this by having you multiply by a conversion factor that secretly equals exactly 1.
- A rate compares two different kinds of quantities, like distance and time, or price and quantity.
- These questions check careful, step-by-step proportional thinking, not a memorized single formula.
- You'll meet this inside real-world word problems about distance, time, money, speed, or density.
- What a Conversion Factor Is — 1 foot = 12 inches, so 12 inches / 1 foot = 1
- The Cancel-the-Units Method — Given quantity $\times$ (desired unit / given unit) = answer in desired unit
- Multi-Step (Chain) Conversions
- Squared and Cubed Units (Area and Volume) — $1 \text{ unit}^2 = \text{conversion factor}^2$; $1 \text{ unit}^3 = \text{conversion factor}^3$
- Rates as Ratios — Rate = quantity / time (or quantity / another quantity)
- Converting a Rate (Two Units at Once)
- Which Direction to Convert

Common Mistakes: Setting up the conversion factor upside down so the starting unit doesn't actually cancel — check units first. ⚠ Forgetting to square or cube the conversion factor for area or volume — always raise the whole factor to that power.

A car travels at a constant speed and covers 108 kilometers in 2 hours. A road sign ahead gives the remaining distance in miles, where 1 mile = 1.6 kilometers. What is the car's speed in miles per hour?

Formula / Rule Used: Find the rate (distance/time), then convert the distance unit.

Work through it, then advance for the full step-by-step solution.

Step 1: First find the speed in kilometers per hour: speed = distance / time = 108 km / 2 hours.

Step 2: Divide: $108 / 2 = 54$, so the car travels at 54 kilometers per hour.

Step 3: Now convert kilometers to miles using 1 mile = 1.6 kilometers, so 1 km = $(1/1.6)$ miles.

Step 4: Multiply: 54 km/hour x (1 mile / 1.6 km).

Step 5: The km unit cancels, leaving miles per hour.

Step 6: Divide: $54 / 1.6 = 33.75$.

Step 7: So the car's speed is 33.75 miles per hour.

Answer: 33.75 miles per hour

Tip: When a problem needs both a rate calculation AND a unit conversion, find the rate first, then convert its unit.

Trap: Since a mile is a bigger distance than a kilometer, converting km to miles should shrink the number — dividing, not multiplying, by 1.6.



Tempting Wrong Answers — and Exactly What Went Wrong

54 miles per hour

Found the correct speed in km/h but forgot to convert kilometers to miles at all.

86.4 miles per hour

Multiplied 54 by 1.6 instead of dividing, going the wrong direction on the conversion.

27 miles per hour

Divided the already-found speed by 2 again, confusing the mile conversion factor with the time value.

Shortcut: You could convert 108 km to miles first ($108/1.6 = 67.5$ miles), then divide by 2 hours — either order reaches 33.75 mph.

Desmos Tip: Type $108/2/1.6$ into Desmos in one line to confirm the speed is 33.75 miles per hour.

What It Is & Core Rules

- Statistics questions ask you to summarize a whole list of numbers with just one or two key values.
- 'Center' means a typical, middle-ish value for the data — mean, median, or mode.
- 'Spread' means how bunched together or spread out the data is — range or standard deviation.
- The SAT tests this because real data (grades, prices, heights) is easier to compare using summaries than raw lists.
- Most questions just want you to calculate one summary value, or predict how it changes when the data changes.
- Mean (Average) — $\text{Mean} = (\text{sum of all values}) / (\text{number of values})$
- Median (Middle Value) — Median = middle value once data is sorted
- Mode and Range — Mode = most frequent value. Range = $\text{max} - \text{min}$.
- Weighted Average — $\text{Weighted avg} = (\text{mean1} \times n1 + \text{mean2} \times n2 + \dots) / (n1 + n2 + \dots)$
- Recovering a Total with $\text{Sum} = \text{Mean} \times n$ — $\text{Sum} = \text{Mean} \times n$
- Standard Deviation (compare only, never calculate)
- Effect of Outliers and Skew — Right-skew: $\text{mean} > \text{median}$. Left-skew: $\text{mean} < \text{median}$. Symmetric: $\text{mean} \approx \text{median}$.

Common Mistakes: Finding the median without sorting the data first, which gives a completely wrong middle value. ⚠️ Averaging group means directly instead of weighting each by its own group size.

A cycling team based in Lyon has 6 riders, and their mean finishing time in a time trial was 15 minutes. One rider got a flat tire and finished in 33 minutes. If that rider's time is excluded from the data, what is the mean finishing time, in minutes, of the remaining 5 riders?

Formula / Rule Used: $\text{Sum} = \text{Mean} \times n$

Work through it, then advance for the full step-by-step solution.

Step 1: Use $\text{Sum} = \text{Mean} \times n$ to turn the original mean back into a total: $\text{sum} = 15 \times 6 = 90$ minutes.

Step 2: This 90 minutes is the combined finishing time of all 6 riders.

Step 3: Remove the excluded rider's time from that total: $90 - 33 = 57$ minutes.

Step 4: There are now only 5 riders left in the data set, so the new n is 5.

Step 5: Divide the new sum by the new count: $\text{mean} = 57 / 5$.

Step 6: New mean = 11.4 minutes.

Answer: 11.4 minutes

Tip: When one value is removed and you need a new mean, first turn the old mean back into a sum with $\text{Sum} = \text{Mean} \times n$.

Trap: It's tempting to subtract 33 from the old mean of 15 directly instead of working through the total sum first.

Tempting Wrong Answers — and Exactly What Went Wrong

-18

Subtracted the removed rider's time straight from the mean, $15 - 33$, instead of adjusting the total sum.

9.5

Divided the new sum of 57 by the original count of 6 instead of the new count of 5.

18

Divided the original total of 90 by the new count of 5 but forgot to subtract the removed rider's time first.

Shortcut: $\text{Sum} = \text{Mean} \times n$ turns this into simple arithmetic — you never need to know the other 5 individual times.

Desmos Tip: Type $(15 \times 6 - 33) / 5$ into Desmos as one line to instantly get the new mean, 11.4.

What It Is & Core Rules

- Data displays are pictures that organize a list of numbers so patterns are easy to see at a glance.
- The SAT shows the same kind of data as histograms, dot plots, or box plots and expects you to read each one.
- Each display highlights something different: dot plots show every exact value, histograms show grouped counts, box plots show spread.
- Most questions just ask you to read the picture correctly, not to calculate anything new from scratch.
- You will never need to draw one of these displays yourself, only interpret one that's already given.
- Histograms — bar height = frequency (count) in that interval
- Dot Plots
- Box Plots and the Five-Number Summary — Min, Q1, Median, Q3, Max
- Interquartile Range (IQR) — $IQR = Q3 - Q1$
- Finding the Median from a List or Dot Plot
- Reading Skew from Shape — Right-skew: $\text{mean} > \text{median}$. Left-skew: $\text{mean} < \text{median}$.
- Comparing Two Data Displays

Common Mistakes: Confusing a histogram's bars (grouped number ranges) with a bar graph's bars (separate labeled categories). ⚠ Miscounting a dot plot by reading the number line value instead of counting how many dots sit above it.

A dot plot represents the number of hours volunteered last month by 7 students at a community center in Dublin. Six of the values are known: 2, 3, 3, 4, 5, and 5 hours. The seventh student's number of hours, v , is unknown. If the mean number of volunteer hours for all 7 students is 4, what is the value of v ?

Formula / Rule Used: Sum of all values = Mean $\times$ number of values

Work through it, then advance for the full step-by-step solution.

Step 1: Recall that $\text{Mean} = \text{Sum} / n$, so $\text{Sum} = \text{Mean} \times n$.

Step 2: There are 7 students total and the mean is 4, so the sum of all 7 values must be $4 \times 7 = 28$.

Step 3: Add up the six known values: $2 + 3 + 3 + 4 + 5 + 5$.

Step 4: Simplify: $2+3+3+4+5+5 = 22$.

Step 5: The unknown value v must make the total reach 28, so $v = 28 - 22$.

Step 6: Subtract: $v = 6$.

Answer: $v = 6$ hours

Tip: Whenever one value is missing but you know the mean, use $\text{Sum} = \text{Mean} \times n$ to find the total first, then subtract the known values.

Trap: It's tempting to average the six known values directly to estimate v , but that ignores that the TRUE mean of all seven students must equal 4.



Tempting Wrong Answers — and Exactly What Went Wrong

4

Assumed the missing value must simply equal the given mean, without actually setting up and solving the equation.

2

Used $n = 6$ instead of 7 in $\text{Sum} = \text{Mean} \times n$, forgetting the unknown student is also part of the group, getting target sum 24 and $v = 24 - 22 = 2$.

50

Added the known sum to the target total instead of subtracting, computing $28 + 22 = 50$ instead of $28 - 22$.

Shortcut: $\text{Sum} = \text{Mean} \times n$ is always faster than guessing — it turns any 'missing value' question into a single subtraction.

Desmos Tip: After solving for $v = 6$ by hand, type `mean({2,3,3,4,5,5,6})` in Desmos to confirm it outputs exactly 4.

What It Is & Core Rules

- A scatterplot is a graph with dots showing two number values for each item, one value per axis.
- It's used to see if two things are connected, like hours studied and a test score.
- The SAT tests this because real data is messy, so you must judge the overall trend, not one dot.
- A line of best fit is a single straight line drawn through the cloud of dots to summarize the trend.
- Scatterplot questions combine reading a graph with basic algebra, like slope-intercept substitution.
- Line of Best Fit — $y = mx + b$
- Types of Association
- Interpreting Slope and Intercept in Context — slope = change in y per 1 unit of x
- Residuals — residual = actual y - predicted y
- Points On the Line vs. Actual Data Points
- Nonlinear Models — quadratic: $y = ax^2 + bx + c$; exponential: $y = a \cdot b^x$
- Using the Model's Equation to Predict or Solve — substitute a known x or y into the equation

Common Mistakes: Treating an actual scattered dot as if it must sit exactly on the line of best fit. ⚠ Flipping the sign of a residual by subtracting predicted minus actual instead of actual minus predicted.

A scatterplot from a sleep study of Belgian students plots hours of sleep the night before on the x-axis (ranging from about 4 to 10) against reaction time in milliseconds on the y-axis (ranging from about 180 to 280). The plotted points trend downward and stay fairly close to a straight line. The line of best fit is $y = -15x + 340$, where x is hours of sleep and y is reaction time in milliseconds. According to this line of best fit, how many hours of sleep does the model predict would produce a reaction time of exactly 190 milliseconds?

Formula / Rule Used: Line of best fit solved in reverse: substitute y and solve for x .

Work through it, then advance for the full step-by-step solution.

Step 1: Normally you'd plug in x to find y , but here you're given $y = 190$ and must find x instead.

Step 2: Substitute $y = 190$ into the equation: $190 = -15x + 340$.

Step 3: Subtract 340 from both sides: $190 - 340 = -15x$.

Step 4: Simplify the left side: $-150 = -15x$.

Step 5: Divide both sides by -15: $x = -150 / -15$.

Step 6: $x = 10$.

Answer: 10 hours

Tip: When a question gives you a y -value and asks for x , substitute y into the equation first, then solve for x algebraically.

Trap: It's tempting to keep isolating x the same way you would for a normal x -to- y prediction, forgetting the roles are reversed here.

Tempting Wrong Answers — and Exactly What Went Wrong

-10 hours

Made a sign error dividing -150 by -15, forgetting a negative divided by a negative gives a positive result.

-12.67 hours

Divided 190 directly by -15 without first subtracting 340 from both sides.

10.7 hours

Misread the slope in the equation as -14 instead of -15, giving a slightly off value.

Shortcut: Once rearranged, you can isolate x in one clean step: $(340 - 190) / 15 = 150 / 15 = 10$.

Desmos Tip: Type $y = -15x + 340$ in Desmos, then trace the line to find where $y = 190$, confirming $x = 10$.

What It Is & Core Rules

- This skill checks whether a survey or study's results can actually be trusted.
- A 'population' is the entire group you want to learn something about.
- A 'sample' is the smaller group that was actually surveyed or tested.
- The SAT tests this because polls, studies, and news reports rely on sampling every day.
- You'll judge whether a sampling method was fair, or work with margin of error numbers.
- Population vs. Sample
- Random Selection Is What Makes a Sample Valid
- Sampling Bias
- Self-Selection and Nonresponse Bias
- Margin of Error — true population value $\approx$ sample statistic $\pm$ margin of error
- Building the Interval From Margin of Error — interval = sample estimate $\pm$ margin of error
- Generalizing Results and Correlation vs. Causation

Common Mistakes: Assuming a bigger sample size fixes a biased sampling method instead of the method itself. ⚠ Thinking a smaller margin of error means a smaller sample, when it's actually the opposite.

In a city of 12,000 residents, a transportation study estimates that between 3,120 and 3,480 residents bike to work. What is the margin of error for this estimate, expressed as a percentage of the city's population?

Formula / Rule Used: Margin of error (count) = (upper bound - lower bound) / 2, then convert to a percent of the population.

Work through it, then advance for the full step-by-step solution.

Step 1: Find the width of the interval in people: $3,480 - 3,120$.

Step 2: Subtract: $3,480 - 3,120 = 360$.

Step 3: The margin of error in people is half the width, since the estimate is the midpoint: $360 / 2$.

Step 4: Divide: $360 / 2 = 180$ people.

Step 5: Convert this to a percent of the population by dividing by 12,000: $180 / 12,000$.

Step 6: Divide: $180 / 12,000 = 0.015$.

Step 7: Multiply by 100 to get a percent: $0.015 \times 100 = 1.5$.

Step 8: The margin of error is 1.5% of the population.

Answer: 1.5%

Tip: When bounds are given as raw counts, find the margin of error in people first, then convert it to a percent of the total population.

Trap: Forgetting to divide the interval's width by 2 gives the full range of people (360), not the actual margin of error.

Tempting Wrong Answers — and Exactly What Went Wrong

3%

Used the full width of 360 people without dividing by 2, then converted that full width to a percent.

180%

Found the correct margin of error in people (180) but forgot to convert it into a percent of the population.

2.85%

Divided the margin of 180 by the midpoint estimate of 3,300 instead of by the total population of 12,000.

Shortcut: Notice 360 people halves easily to 180, then divide by 12,000 by simplifying to 18/1,200 mentally.

Desmos Tip: Type $((3480-3120)/2)/12000*100$ directly into Desmos as one line to get the margin of error percent in a single step.



PART 3

Advanced Math

Functions & Notation · Composite & Inverse Functions · Transformations · Polynomial Operations & Factoring · Polynomial Division & Rational Expressions · Quadratics · Radicals, Rational Exponents & Exponential Functions · Radical & Rational Equations

What It Is & Core Rules

- A function is a rule that takes an input and gives back exactly one output for it.
- Function notation $f(x)$ is just a labeled way to write 'the output of rule f when the input is x .'
- The SAT uses functions to test whether you can plug in, read graphs, and track inputs vs. outputs.
- You'll meet functions as equations, as graphs, as tables of values, and as word-problem descriptions.
- Domain and range just describe which inputs are allowed and which outputs actually happen.
- Function Notation — $y = f(x)$
- Evaluating a Function — $f(a)$ = substitute a for every x
- The Vertical Line Test
- Domain and Range — Domain = allowed x -values. Range = resulting y -values.
- Domain Restrictions — Denominator $\neq 0$. Radicand under an even root ≥ 0 .
- Piecewise Functions — $f(x) = \{\text{expression 1 if condition 1; expression 2 if condition 2}\}$
- Rate of Change Between Two Points — Rate of change = $(f(x_2) - f(x_1)) / (x_2 - x_1)$

Common Mistakes: Treating $f(x)$ like multiplication (f times x) instead of 'the output of f at input x .' ⚠ Substituting the input into only part of the expression instead of every occurrence of x .



A function f is defined by $f(x) = 3x + 2$ if $x < 2$, and $f(x) = x^2 - 1$ if $x \geq 2$. For what value of x is $f(x) = 24$?

Formula / Rule Used: Piecewise functions: solve each piece, then check its condition.

Work through it, then advance for the full step-by-step solution.



Step 1: Since 24 is a fairly large output, try the $x \geq 2$ piece first: $x^2 - 1 = 24$.

Step 2: Add 1 to both sides: $x^2 = 25$.

Step 3: Take the square root of both sides: $x = 5$ or $x = -5$.

Step 4: Check each against the condition $x \geq 2$: $x = 5$ satisfies it, but $x = -5$ does not.

Step 5: Also check the other piece to be thorough: $3x + 2 = 24$ gives $x = 22/3$.

Step 6: But $22/3$ is about 7.33, which does NOT satisfy that piece's condition $x < 2$, so it's rejected.

Step 7: The only value that works in its own piece is $x = 5$.

Answer: $x = 5$

Tip: Guess which piece the output likely belongs to based on its size, solve that piece, then always verify the condition.

Trap: It's tempting to report both $x = 5$ and $x = -5$ without checking that -5 fails the required condition $x \geq 2$.



Tempting Wrong Answers — and Exactly What Went Wrong

$x = 5$ or $x = -5$

Solved $x^2 = 25$ correctly but forgot to reject $x = -5$, which fails the piece's condition $x \geq 2$.

$x = 22/3$

Solved the first piece ($3x + 2 = 24$) but never checked that $22/3$ fails that piece's own condition $x < 2$.

$x = -5$

Took only the negative square root of 25 without checking it against the required condition $x \geq 2$.

Shortcut: Starting with the piece whose condition matches the output's size (a large positive output suggests the squared piece) saves time over checking both blindly.

Desmos Tip: Graph $y = 3x + 2$ restricted to $x < 2$ and $y = x^2 - 1$ restricted to $x \geq 2$ in Desmos, then trace to find where $y = 24$.

What It Is & Core Rules

- A composite function is what you get when you plug one whole function inside another one.
- Writing $f(g(x))$ means: find $g(x)$ first, then use that answer as the new input for f .
- An inverse function is a function that completely undoes what the original function does.
- The SAT tests these to see if you can carefully track which expression substitutes into which.
- Both ideas show up as plain equations, tables of values, or graphs — not just algebra to solve.
- Composite Function Notation — $(f \circ g)(x) = f(g(x))$
- Evaluating a Composite at a Number — Work Inside Out
- Building the Composite Expression First — $f(g(x))$ = substitute $g(x)$ for every x in $f(x)$
- Expanding Multi-Term Substitutions Correctly — $(a + b)^2 = a^2 + 2ab + b^2$
- What an Inverse Function Is — $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$
- Finding an Inverse Algebraically — Steps: 1) $y = f(x)$ 2) swap x and y 3) solve for y 4) $y = f^{-1}(x)$
- Inverse from a Table, Graph, or Reflection — $f^{-1}(b) = a$ when $f(a) = b$

Common Mistakes: Evaluating the outer function first instead of the inner one, which reverses the whole answer. ⚠ Distributing an exponent across a substituted expression instead of expanding it fully.



If $h(x - 3) = 2x^2 + x - 4$, what is $h(5)$?

Formula / Rule Used: Disguised composite: match the inside expression to the target input.

Work through it, then advance for the full step-by-step solution.



Step 1: h is defined in terms of $(x - 3)$, not x directly, so $h(5)$ needs $x - 3$ to equal 5.

Step 2: Solve $x - 3 = 5$ for x : $x = 8$.

Step 3: Substitute $x = 8$ into the given right-hand expression: $2(8)^2 + 8 - 4$.

Step 4: Square first: $8^2 = 64$, so $2(64) = 128$.

Step 5: Add and subtract left to right: $128 + 8 = 136$, then $136 - 4 = 132$.

Answer: 132

Tip: When you see $h(\text{something} - \text{number})$, set that something equal to the target input and solve for x first.

Trap: It's tempting to plug $x = 5$ straight into the expression, skipping the step of solving $x - 3 = 5$.

Tempting Wrong Answers — and Exactly What Went Wrong

51

Plugged $x = 5$ directly into $2x^2 + x - 4$ instead of first solving $x - 3 = 5$ for $x = 8$.

124

Found $x = 8$ correctly, but mis-sequenced the last step as $128 - 8 + 4$ instead of $128 + 8 - 4$.

6

Solved $x - 3 = 5$ with a sign error, getting $x = 2$ instead of $x = 8$, then used that wrong x .

Shortcut: None beyond solving for x first — once $x = 8$ is found, direct substitution is already fastest.

Desmos Tip: Type $y1 = 2x^2 + x - 4$ in Desmos and trace to $x = 8$ to confirm $y1 = 132$, matching $h(5)$.

What It Is & Core Rules

- A transformation takes a function's graph and shifts, flips, stretches, or squishes it, without changing its basic shape.
- You start from a 'parent' graph, like $y = f(x)$, and a small change to the equation moves the whole picture.
- The SAT tests whether you can predict exactly how a graph moves just by reading numbers added or multiplied in the equation.
- Every transformation question is really asking: does this change move the graph up/down, left/right, flip it, or resize it?
- You'll see this tested with equations, with graphs you compare by eye, and with word problems describing a shifted model.
- Vertical Shift (moves up or down) — $y = f(x) + k$
- Horizontal Shift (moves left or right) — $y = f(x + h)$
- Why Horizontal Shifts Flip Direction
- Reflections (flips the graph) — $y = -f(x)$ flips over the x-axis. $y = f(-x)$ flips over the y-axis.
- Vertical Stretch and Compression — $y = a \cdot f(x)$
- Horizontal Stretch and Compression — $y = f(bx)$
- Combining Several Transformations at Once — $y = a \cdot f(x - h) + k$

Common Mistakes: Assuming $f(x + 3)$ shifts right — it actually shifts LEFT; always check the sign rule for horizontal shifts. ⚠ Mixing up inside changes (affect x , do the opposite of what you'd guess) with outside changes (affect y , match your guess).

The graph of $y = f(x)$ contains the point $(3, 5)$. The function $g(x) = 2f(x - 4) + 1$ is a transformation of f . What point corresponds to $(3, 5)$ on the graph of g ?

Formula / Rule Used: Combined transformation: $y = a \cdot f(x - h) + k$

Work through it, then advance for the full step-by-step solution.

Step 1: $g(x) = 2f(x - 4) + 1$ combines a shift right 4, a vertical stretch by 2, and a shift up 1.

Step 2: First find the new x-coordinate: g uses f at input $(x - 4)$, so set $x - 4 = 3$.

Step 3: Add 4 to both sides: $x = 3 + 4 = 7$.

Step 4: This tells us the original point moves to $x = 7$ on the graph of g.

Step 5: Next find the new y-coordinate by applying the outside operations to $y = 5$.

Step 6: Multiply first: $2 * 5 = 10$.

Step 7: Then add the vertical shift: $10 + 1 = 11$.

Step 8: So the corresponding point on g is $(7, 11)$.

Answer: $(7, 11)$

Tip: Handle the inside (solve $x - 4 = 3$ for the new x) and the outside (compute $2(5) + 1$ for the new y) as two separate steps.

Trap: A common mistake is doing the addition before the multiplication on the y-value, or solving $x - 4 = 3$ backwards.

Tempting Wrong Answers — and Exactly What Went Wrong

(7, 10)

Forgot to add the final +1, stopping after $2(5) = 10$.

(-1, 11)

Solved $x - 4 = 3$ incorrectly as $x = 3 - 4 = -1$ instead of $x = 3 + 4 = 7$.

(7, 12)

Added 1 before multiplying by 2, computing $2(5 + 1) = 12$ instead of $2(5) + 1$.

Shortcut: None needed — solving the inside and outside separately is already the fastest method for a combined transformation.

Desmos Tip: Define $f(x)$ as any curve through (3, 5), type $g(x) = 2f(x-4)+1$, and trace g at $x = 7$ to confirm $y = 11$.

What It Is & Core Rules

- A polynomial is an expression made of terms with whole-number exponents, like $3x^2 + 5x - 7$.
- Polynomial operations means adding, subtracting, or multiplying these expressions together.
- Special products are multiplication patterns, like $(a + b)^2$, that appear so often they're worth memorizing.
- Memorizing the patterns lets you expand instantly instead of multiplying every term by every term.
- The SAT tests this because these patterns hide inside harder algebra, geometry, and factoring problems.
- Polynomial Vocabulary — $a_n \cdot x^n + \dots + a_1 x + a_0$
- Adding Polynomials
- Subtracting Polynomials
- Multiplying Polynomials / FOIL — $(a+b)(c+d) = ac + ad + bc + bd$
- Square of a Sum — $(a + b)^2 = a^2 + 2ab + b^2$
- Square of a Difference — $(a - b)^2 = a^2 - 2ab + b^2$
- Difference of Squares — $(a + b)(a - b) = a^2 - b^2$

Common Mistakes: Writing $(a + b)^2$ as $a^2 + b^2$ — the middle term $2ab$ is real and cannot be skipped. ⚠ Forgetting to flip the sign of every term when subtracting one polynomial from another.



The expression $x^2 + 14x + k$ is a perfect square trinomial, equal to $(x + h)^2$ for some positive constants h and k . What is the value of k ?

Formula / Rule Used: Perfect square trinomial pattern: $(x + h)^2 = x^2 + 2hx + h^2$.

Work through it, then advance for the full step-by-step solution.



Step 1: Compare $x^2 + 14x + k$ to the pattern $x^2 + 2hx + h^2$.

Step 2: Match the middle terms: the coefficient of x must satisfy $2h = 14$.

Step 3: Solve for h by dividing both sides by 2: $h = 7$.

Step 4: The constant term of a perfect square trinomial is always h^2 , so $k = h^2$.

Step 5: Substitute $h = 7$: $k = 7^2 = 49$.

Answer: $k = 49$

Tip: Match the middle-term coefficient to $2h$ first — once you know h , k is simply h squared.

Trap: A common mistake is reporting k as equal to h itself (7) instead of squaring h to get 49.



Tempting Wrong Answers — and Exactly What Went Wrong

$k = 7$

Correctly found $h = 7$ but reported it directly as k instead of squaring it.

$k = 14$

Confused the middle-term coefficient 14 with the value of h^2 and reported it unchanged.

$k = 196$

Squared the middle-term coefficient 14 directly instead of first solving for h and then squaring h .

Shortcut: Since the trinomial is guaranteed to be a perfect square, jump straight to $k = \text{coefficient of } x / 2^2$ without expanding anything.

Desmos Tip: Type $y = x^2 + 14x + 49$ and confirm it touches the x -axis at exactly one point, showing the trinomial is a perfect square when $k = 49$.

What It Is & Core Rules

- Factoring takes an expanded expression, like $x^2 + 5x + 6$, and rewrites it as a product of factors.
- It's the exact reverse of multiplying (distributing) two expressions together.
- The SAT tests factoring because it's usually the fastest way to solve a quadratic or cubic equation.
- Recognizing the right pattern instantly (GCF, trinomial, difference of squares) saves real time on the test.
- Factoring also hides inside other topics, like simplifying rational expressions and reading a graph's roots.
- Factor Out the GCF First — $\text{GCF} \times (\text{remaining terms})$
- Factoring $x^2 + bx + c$ (leading coefficient 1) — Find r, s where $r \times s = c$ and $r + s = b$, then $(x + r)(x + s)$
- The AC Method for $ax^2 + bx + c$ (a is not 1) — Find two numbers that multiply to $a \times c$ and add to b
- Difference of Squares — $a^2 - b^2 = (a - b)(a + b)$
- Perfect Square Trinomial — $a^2 \pm 2ab + b^2 = (a \pm b)^2$
- Factoring by Grouping (four-term polynomials, including cubics) — Group the first two terms and the last two terms separately
- Multiplicity of Roots — The exponent on a factor = that root's multiplicity

Common Mistakes: Skipping the GCF check first, which makes every later factoring step messier than it needs to be. ⚠ Stopping after one factoring step instead of checking whether a resulting factor can be factored again.

What is the complete factored form of $x^4 - 81$?

Formula / Rule Used: Difference of squares, applied twice (a disguised form).

Work through it, then advance for the full step-by-step solution.

Step 1: Recognize that $x^4 - 81$ is itself a difference of squares: $(x^2)^2 - 9^2$.

Step 2: Apply the difference of squares pattern once: $x^4 - 81 = (x^2 - 9)(x^2 + 9)$.

Step 3: Look at each new factor separately to check whether either can be factored again.

Step 4: The factor $x^2 + 9$ is a SUM of squares, which cannot be factored over the real numbers.

Step 5: The factor $x^2 - 9$ is itself another difference of squares: $(x - 3)(x + 3)$.

Step 6: Replace $x^2 - 9$ with its factored form, keeping $x^2 + 9$ exactly as it is.

Step 7: The fully factored form is $(x - 3)(x + 3)(x^2 + 9)$.

Answer: $(x - 3)(x + 3)(x^2 + 9)$

Tip: After one round of difference of squares, always check whether either new factor is itself a difference of squares.

Trap: It's tempting to also try to factor $x^2 + 9$ — but a SUM of squares never factors over the real numbers, so stop there.

Tempting Wrong Answers — and Exactly What Went Wrong

$$(x - 3)(x + 3)(x - 3)(x + 3)$$

Mistakenly treated $x^2 + 9$ as another difference of squares and force-factored it, even though a sum of squares can't be factored.

$$(x^2 - 9)(x^2 + 9)$$

Stopped after the first factoring step without checking whether the factor $x^2 - 9$ could be factored again.

$$(x - 9)(x + 9)$$

Incorrectly took the square root of x^4 as x instead of x^2 , misapplying the pattern to the wrong power.

Shortcut: None faster here — the two-step process of factoring, then checking each new piece again, is the reliable way to fully factor a disguised form.

Desmos Tip: Type $y1 = x^4 - 81$ and $y2 = (x - 3)(x + 3)(x^2 + 9)$ into Desmos to confirm the two graphs match exactly.

What It Is & Core Rules

- Dividing polynomials means splitting one polynomial expression by another, much like long division with numbers.
- A rational expression is simply a fraction where the top, the bottom, or both are polynomials.
- This skill combines factoring with fraction rules, so both must be solid before you start.
- Any x-value that would make a denominator zero must always be excluded from the answer.
- These ideas show up together: simplifying, combining, dividing, and solving equations built from rational expressions.
- Dividing a Polynomial by a Monomial — $(a+b+c)/d = a/d + b/d + c/d$
- Polynomial Long Division
- Simplifying Rational Expressions
- Multiplying and Dividing Rational Expressions — $(a/b) \times (c/d) = ac/bd$
- Adding and Subtracting Rational Expressions — $a/b + c/d = (ad + bc)/bd$
- Solving Rational Equations
- Complex Fractions

Common Mistakes: Canceling a single term instead of a whole factor, like crossing out the x in $(x+3)/x$. ⚠ Simplifying a rational expression correctly but forgetting to state the x-value it excludes.



Write $\frac{2}{x+3} + \frac{1}{x^2-9}$ as a single fraction in simplest form.

Formula / Rule Used: Adding rational expressions: factor first, find the LCD, then combine numerators.

Work through it, then advance for the full step-by-step solution.



Step 1: Factor the second denominator: $x^2 - 9 = (x - 3)(x + 3)$.

Step 2: Compare the two denominators, $(x+3)$ and $(x-3)(x+3)$ — the LCD is $(x-3)(x+3)$.

Step 3: Rewrite the first fraction using the LCD: $\frac{2}{(x+3)} = \frac{2(x-3)}{(x-3)(x+3)}$.

Step 4: Multiply out that new numerator: $2(x-3) = 2x - 6$.

Step 5: The second fraction already uses the LCD, so add the numerators: $(2x-6) + 1 = 2x - 5$.

Step 6: Combine everything over the shared denominator: $\frac{2x-5}{(x-3)(x+3)}$.

Step 7: Multiply the denominator back out if needed: $\frac{2x - 5}{x^2 - 9}$.

Answer: $\frac{2x - 5}{x^2 - 9}$, where $x \neq 3$ and $x \neq -3$

Tip: Whenever a denominator can be factored, factor it first — it usually reveals the true LCD hiding underneath.

Trap: Treating $(x+3)$ and $(x^2 - 9)$ as unrelated and multiplying them together as the LCD creates unnecessary extra factors.

Tempting Wrong Answers — and Exactly What Went Wrong

$$\frac{3}{x^2 - 9}$$

Added the numerators 2 and 1 directly to get 3 without first rewriting $\frac{2}{x+3}$ over the LCD.

$$\frac{(2x - 5)}{(x + 3)^2(x - 3)}$$

Multiplied the two original denominators together as if they shared no common factor, creating an oversized LCD.

$$\frac{(2x - 6)}{x^2 - 9}$$

Rewrote the first fraction correctly but forgot to add the second numerator's 1 to the top.

Shortcut: None faster — spotting the factored relationship between the two denominators is itself the shortcut here.

Desmos Tip: Type $y1 = \frac{2}{x + 3} + \frac{1}{x^2 - 9}$ and $y2 = \frac{(2x - 5)}{x^2 - 9}$ in Desmos; identical graphs confirm the combined fraction is correct.

What It Is & Core Rules

- A quadratic equation is any equation where the highest power of x is x^2 , like $ax^2 + bx + c = 0$.
- Solving it means finding every x -value that makes the equation true.
- Most quadratics have two solutions, but some have exactly one, and some have none at all in real numbers.
- The SAT tests four different solving methods, plus a shortcut, the discriminant, that predicts how many solutions exist.
- These skills reappear constantly, since many word problems and later topics ultimately reduce to solving a quadratic.
- Solving by Factoring — $ax^2 + bx + c = 0 \rightarrow$ factor $\rightarrow$ set each factor to 0
- Solving by Taking Square Roots — If $x^2 = k$, then $x = \pm \sqrt{k}$
- Solving by Completing the Square — $x^2 + bx \rightarrow (x + b/2)^2 - (b/2)^2$
- The Quadratic Formula — $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- The Discriminant — $D = b^2 - 4ac$
- Choosing the Best Method

Common Mistakes: Forgetting the $\pm$ sign when taking a square root, which throws away one of the two valid solutions. ⚠ Plugging numbers into the discriminant or quadratic formula before rearranging the equation into standard form.



One solution of $x^2 - 4x + k = 0$ is $x = 6$. What is the other solution?

Formula / Rule Used: Substitute a known root to find a missing constant, then factor to find the other root.

Work through it, then advance for the full step-by-step solution.

Step 1: Since $x = 6$ is a solution, substitute it into the equation to solve for the missing constant, k : $6^2 - 4(6) + k = 0$.

Step 2: Compute the numbers: $36 - 24 + k = 0$.

Step 3: Simplify: $12 + k = 0$.

Step 4: Solve for k : $k = -12$.

Step 5: Now the full equation is $x^2 - 4x - 12 = 0$.

Step 6: Factor this: find two numbers that multiply to -12 and add to -4 , which are -6 and 2 .

Step 7: Write the factored form: $(x - 6)(x + 2) = 0$, so $x = 6$ or $x = -2$.

Step 8: Since 6 is already the given solution, the other solution is $x = -2$.

Answer: $x = -2$

Tip: When one solution is given, plug it in first to solve for any unknown constant before finding the other solution.

Trap: It's tempting to stop once you've solved for k , forgetting the question actually asks for the OTHER value of x .

Tempting Wrong Answers — and Exactly What Went Wrong

k = -12

Correctly solved for the missing constant but reported k instead of the other value of x the question asked for.

x = 2

Made a sign error factoring -12 and -4, using the pair 6 and -2 in the wrong factor, flipping the sign of the answer.

x = -6

Reused the given root, 6, with a flipped sign as the 'other' solution instead of actually factoring the equation.

Shortcut: None faster here — substituting the known root to find k, then factoring, is the reliable two-step path.

Desmos Tip: Type $y = x^2 - 4x - 12$ into Desmos and use the zero/root feature to confirm the two x-intercepts are 6 and -2.

What It Is & Core Rules

- A quadratic equation always contains an x^2 term, and its graph is a curved shape called a parabola.
- A parabola always opens either upward (like a smile) or downward (like a frown), never sideways.
- The SAT writes the same parabola in three different equation forms, and each form reveals a different feature instantly.
- Roots, also called x-intercepts or zeros, are the x-values where the parabola crosses the x-axis.
- This skill tests whether you can read a parabola's vertex, intercepts, and roots straight from its equation, without graphing it.
- Standard Form — $y = ax^2 + bx + c$
- Factored Form — $y = a(x - p)(x - q)$
- Vertex Form — $y = a(x - h)^2 + k$
- Completing the Square — Converts standard form into vertex form
- Vieta's Formulas (Sum and Product of Roots) — Sum of roots = $-b/a$. Product of roots = c/a .
- The Discriminant and the Graph — Discriminant = $b^2 - 4ac$
- Quadratic Word Problems

Common Mistakes: Forgetting that a downward parabola ($a < 0$) has a maximum, not a minimum — always check the sign of a first. ⚠ Reading the vertex's x-coordinate as $-b/a$ instead of the correct $-b/(2a)$, forgetting to double the denominator.



A parabola has vertex (2, -3) and passes through the point (5, 15). The equation can be written as $y = a(x - 2)^2 - 3$. What is the value of a ?

Formula / Rule Used: Vertex form $y = a(x - h)^2 + k$, solved for a using a known point.

Work through it, then advance for the full step-by-step solution.

Step 1: Since the vertex is (2, -3), the equation has the form $y = a(x - 2)^2 - 3$.

Step 2: The parabola also passes through (5, 15), so substitute $x = 5$ and $y = 15$.

Step 3: This gives $15 = a(5 - 2)^2 - 3$.

Step 4: Simplify inside the parentheses first: $15 = a(3)^2 - 3$, which is $15 = 9a - 3$.

Step 5: Add 3 to both sides: $18 = 9a$.

Step 6: Divide both sides by 9: $a = 2$.

Answer: $a = 2$

Tip: When you're given a vertex and one more point, plug that point straight into vertex form and solve for the leftover constant.

Trap: It's easy to forget to square (5 - 2) before multiplying by a , treating the equation as $15 = 3a - 3$ instead of $15 = 9a - 3$.



Tempting Wrong Answers — and Exactly What Went Wrong

$a = 6$

Skipped squaring $(5-2)$, using $15 = 3a - 3$ instead of $15 = 9a - 3$, giving $18 = 3a$.

$a = -2$

Made a sign error moving the -3 , writing $15 = 9a + 3$ instead of $15 = 9a - 3$.

$a = 1.8$

Divided 18 by 10 instead of by 9, a simple arithmetic slip.

Shortcut: None faster here — substituting the known point into vertex form is already the most direct route to a .

Desmos Tip: In Desmos, graph $y = a(x - 2)^2 - 3$ with a slider for a , and adjust it until the curve passes through $(5,15)$ to confirm $a = 2$.

What It Is & Core Rules

- A radical, like a square root, and a rational exponent, like $x^{1/2}$, are two ways to write the exact same thing.
- This skill is about simplifying and combining these expressions once a variable is involved, not just plain numbers.
- The SAT hides radicals and fractional exponents inside bigger algebra and function questions to test this fluency.
- You need to convert between the two forms, simplify, combine, and clear radicals from denominators confidently.
- Radical-Exponent Conversion (with Variables) — n -th root of $(x^m) = x^{m/n}$
- Properties of Radicals — $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$ and $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$
- Simplifying Radicals with Variables
- Adding and Subtracting Radicals — $a\sqrt{c} + b\sqrt{c} = (a + b)\sqrt{c}$
- Rationalizing Denominators — Single radical: multiply by that radical. Binomial: multiply by the conjugate.
- Exponent Rules with Rational Exponents — $x^{m/n} \cdot x^{p/n} = x^{(m+p)/n}$

Common Mistakes: Treating $\sqrt{x+y}$ as $\sqrt{x} + \sqrt{y}$ — a radical never distributes over addition or subtraction. ⚠️ Trying to add or subtract radicals before simplifying them, missing that they actually match.



Simplify and combine the expression $\sqrt{(45x)} + \sqrt{(20x)}$ for $x > 0$.

Formula / Rule Used: Simplify each radical first, then combine any resulting like radicals.

Work through it, then advance for the full step-by-step solution.



Step 1: Simplify the first radical: $45x = 9 \cdot 5x$, so $\sqrt{45x} = \sqrt{9 \cdot 5x} = 3 \cdot \sqrt{5x}$.

Step 2: Simplify the second radical: $20x = 4 \cdot 5x$, so $\sqrt{20x} = \sqrt{4 \cdot 5x} = 2 \cdot \sqrt{5x}$.

Step 3: Rewrite the original expression using these simplified forms: $3 \cdot \sqrt{5x} + 2 \cdot \sqrt{5x}$.

Step 4: Notice both terms now share the exact same radicand, $5x$, so they can be combined.

Step 5: Add the outside coefficients only, leaving the radical untouched: $3 + 2 = 5$.

Step 6: Write the combined result: $5 \cdot \sqrt{5x}$.

Answer: $5 \cdot \sqrt{5x}$

Tip: Whenever two radicals look unrelated, simplify each one first — matching radicands often only appear afterward.

Trap: Trying to add $\sqrt{45x}$ and $\sqrt{20x}$ directly, since they look different, without simplifying first leads nowhere.



Tempting Wrong Answers — and Exactly What Went Wrong

$\sqrt{(65x)}$

Added the numbers under the two radicals directly, $45 + 20$, instead of simplifying and combining like radicals.

$3 \cdot \sqrt{(5x)}$

Simplified only the first radical correctly and forgot to also simplify and add in the second term.

$6 \cdot \sqrt{(5x)}$

Simplified both radicals correctly but multiplied the coefficients, 3×2 , instead of adding them.

Shortcut: Spotting that 45 and 20 are both multiples of 5 hints early that both radicals will simplify to a multiple of $\sqrt{(5x)}$.

Desmos Tip: Substitute $x = 1$ into $\sqrt{(45x)} + \sqrt{(20x)}$ and into $5 \cdot \sqrt{(5x)}$ as two Desmos lines and confirm the outputs match.

What It Is & Core Rules

- Exponential functions describe a quantity that grows or shrinks by the same PERCENT every time period.
- The general form is $y = a \cdot b^x$, where x is usually time and y is the changing amount.
- This is different from a linear function, which adds the same fixed amount every step instead of multiplying.
- The SAT tests this both as pure algebra (solving equations with x in the exponent) and as real-world stories.
- Deciding whether a situation is growth or decay is always the first move before writing an equation.
- General Exponential Form — $y = a \cdot b^x$
- Turning a Percent Rate into the Base — Growth: $A = P(1 + r)^t$. Decay: $A = P(1 - r)^t$.
- Reading the Basic Graph — y -intercept: $(0, a)$. Horizontal asymptote: $y = 0$.
- Solving by Matching Bases — If $a^m = a^n$ (same base a), then $m = n$.
- Vertical Shifts Move the Asymptote — $y = b^x + k$
- Horizontal Shifts — $y = b^{x-h}$

Common Mistakes: Writing the decay base as a negative number instead of $(1 - r)$ — a decaying base is always positive and less than 1. ⚠ Forgetting a vertical shift moves the asymptote too, and still assuming the asymptote is $y = 0$ after a shift.



A population of fruit flies is modeled by an exponential function $N(t) = N_0 \cdot b^t$, where t is measured in days. At $t = 0$ there are 50 flies, and at $t = 2$ there are 200 flies. Assuming this exponential pattern continues, how many flies will there be at $t = 4$?

Formula / Rule Used: Building an exponential model from two points, then evaluating forward.

Work through it, then advance for the full step-by-step solution.



Step 1: At $t = 0$, N_0 is just the starting value: $N(0) = 50$, so $N_0 = 50$.

Step 2: Use the second point to find b : $N(2) = 50 \cdot b^2 = 200$.

Step 3: Divide both sides by 50 to isolate b^2 : $b^2 = \frac{200}{50} = 4$.

Step 4: Take the square root of both sides (b must be positive for a real population): $b = 2$.

Step 5: The full model is now $N(t) = 50 \cdot 2^t$. Substitute $t = 4$: $N(4) = 50 \cdot 2^4$.

Step 6: Evaluate the power: $2^4 = 16$.

Step 7: Multiply: $N(4) = 50 \cdot 16 = 800$.

Answer: 800 flies

Tip: When you're given two points instead of the growth factor, find b first by dividing, then build the full equation before evaluating.

Trap: A common mistake is assuming the population keeps adding the same fixed amount (150 flies) instead of continuing to multiply.



Tempting Wrong Answers — and Exactly What Went Wrong

350 Assumed linear growth (adding the same 150-fly increase again) instead of continuing the exponential pattern.

400 Found $b^2 = 4$ correctly but only multiplied 200 by $b = 2$ instead of by $b^2 = 4$ for the jump from $t=2$ to $t=4$.

3200 Used the value at $t=2$ (200) as N_0 by mistake, then computed $200 \cdot 2^4$ instead of $50 \cdot 2^4$.

Shortcut: Once you know $b=2$, jump straight from $N(2)=200$ to $N(4)$ by multiplying by $b^2 = 4$, without rewriting the full equation.

Desmos Tip: Type $y = 50 \cdot 2^x$ into Desmos, confirm the curve passes through $(0,50)$ and $(2,200)$, then trace to $x=4$ to confirm $y=800$.

What It Is & Core Rules

- A radical equation has the variable trapped under a square root, cube root, or other root symbol.
- A rational equation has the variable sitting inside a fraction, specifically in the denominator.
- The SAT groups these together because both require the same first move: clear the root or the fraction.
- That clearing step can accidentally create fake solutions, so checking your work matters more here than usual.
- Once the root or fraction is gone, what's left is just a normal equation you already know how to solve.
- Isolate the Radical First
- Raise Both Sides to Match the Index — If n -th root of $x = a$, then $x = a^n$
- Always Check for Extraneous Solutions
- Equations with Two Radicals
- Clear a Rational Equation's Denominators — Multiply every term by the LCD
- Extraneous Solutions in Rational Equations
- Complex Fractions

Common Mistakes: Squaring both sides before fully isolating the radical, which creates messy extra terms. ⚠ Forgetting to check solutions in the original equation, missing a hidden extraneous solution.



Solve for x : $\sqrt{x + 6} = x$.

Formula / Rule Used: Square both sides to eliminate the radical, then check every solution in the original equation.

Work through it, then advance for the full step-by-step solution.

Step 1: The radical is already alone on the left side of the equation.

Step 2: Square both sides: $(\sqrt{x+6})^2 = x^2$.

Step 3: The left side simplifies to $x+6$, so the equation becomes $x+6 = x^2$.

Step 4: Rearrange into standard form by subtracting x and 6 from both sides: $0 = x^2 - x - 6$.

Step 5: Factor the quadratic: $x^2 - x - 6 = (x-3)(x+2)$.

Step 6: Set each factor equal to zero: $x-3=0$ or $x+2=0$, giving $x=3$ or $x=-2$.

Step 7: Check $x=3$ in the original equation: $\sqrt{3+6} = \sqrt{9} = 3$, which matches.

Step 8: Check $x=-2$ in the original equation: $\sqrt{-2+6} = \sqrt{4} = 2$, but 2 does not equal -2 , so $x=-2$ is extraneous and must be rejected.

Answer: $x=3$ ($x=-2$ is extraneous and rejected)

Tip: Whenever squaring a radical equation produces a quadratic, expect two candidate answers and check both in the original equation.

Trap: It's tempting to report both factored solutions without checking that one of them fails in the original equation.

Tempting Wrong Answers — and Exactly What Went Wrong

$x = 3$ and $x = -2$

Solved the quadratic correctly but skipped checking each solution in the original equation, keeping the extraneous root.

$x = -2$

Checked the solutions but mixed up which one satisfies the original equation, keeping the wrong root.

$x = 2$ or $x = -3$

Made a sign error while factoring $x^2 - x - 6$, reversing the signs inside the two factors.

Shortcut: None needed — squaring, factoring, and checking both roots is already the fastest reliable path for this equation type.

Desmos Tip: Graph $y = \sqrt{x + 6}$ and $y = x$ in Desmos; the graphs only cross once, at $x = 3$, confirming the other algebraic root is extraneous.



PART 4

Geometry and Trigonometry

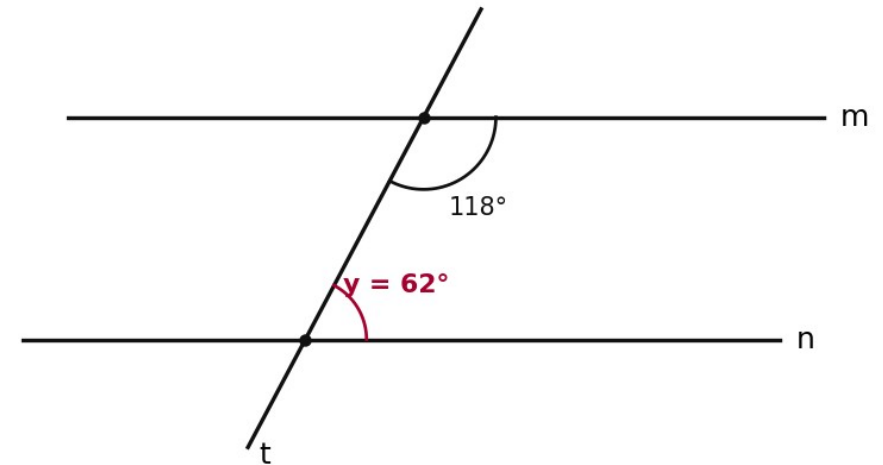
Angles, Parallel Lines & Triangle Fundamentals • Right Triangles & Trigonometry • Radians & the Unit Circle • Circles • Volume, Surface Area & Similarity of Solids

What It Is & Core Rules

- An angle measures how much two rays turn away from each other at a shared starting point.
- Angles are measured in degrees, where a full circle around a point is 360 degrees.
- A triangle is the simplest closed shape, and its angle and side rules never change, no matter the size.
- The SAT tests these rules because they let you find unknown angles and sides without ever measuring anything.
- Parallel lines crossed by a third line create a repeating, predictable pattern of equal and supplementary angles.
- Vertical, Complementary & Supplementary Angles — Vertical angles: equal. Complementary: sum = 90° . Supplementary: sum = 180° .
- Parallel Lines Cut by a Transversal
- Triangle Angle Sum — $A + B + C = 180^\circ$
- Classifying Triangles
- Triangle Inequality Theorem — $a + b > c$ (true for every pair of sides)
- Side-Angle Relationship
- Exterior Angle Theorem — Exterior angle = sum of the two remote interior angles

Common Mistakes: Assuming all eight angles at a transversal crossing are equal, instead of checking which pairs are supplementary. ⚠ Confusing corresponding angles (equal) with same-side interior angles (supplementary) — they look similar but aren't the same.

Line m and line n are parallel, and both are crossed by a transversal line t . At the crossing with line m , one of the angles formed measures 118 degrees. Angle y is the same-side interior angle to it at the crossing with line n (angle y is between lines m and n , on the same side of the transversal as the 118 degree angle). What is the measure of angle y ?



Formula / Rule Used: Same-side interior angles are supplementary (sum to 180 degrees).

Work through it, then advance for the full step-by-step solution.



Step 1: A transversal crossing two parallel lines creates several angle pairs with predictable relationships.

Step 2: Same-side interior angles are the pair that lies between the two parallel lines, on the same side of the transversal.

Step 3: Unlike corresponding or alternate angles, same-side interior angles are NOT equal — they are supplementary.

Step 4: Set up the supplementary equation: $118 + y = 180$.

Step 5: Subtract 118 from both sides to solve for y : $y = 180 - 118$.

Step 6: $y = 62$ degrees.

Answer: 62 degrees

Tip: Before you solve, always identify exactly which of the eight transversal angles you're comparing — same-side interior angles are the one pair that's supplementary, not equal.

Trap: This problem is built to tempt you into treating angle y as equal to 118 degrees, the way corresponding or alternate interior angles would be.



Tempting Wrong Answers — and Exactly What Went Wrong

118 degrees

Treated the same-side interior pair as equal, like corresponding angles, instead of supplementary.

72 degrees

Used complementary angles ($90 - 118$ mishandled) instead of the correct supplementary relationship.

242 degrees

Added $118 + 180 - 56$ style arithmetic error instead of correctly subtracting 118 from 180.

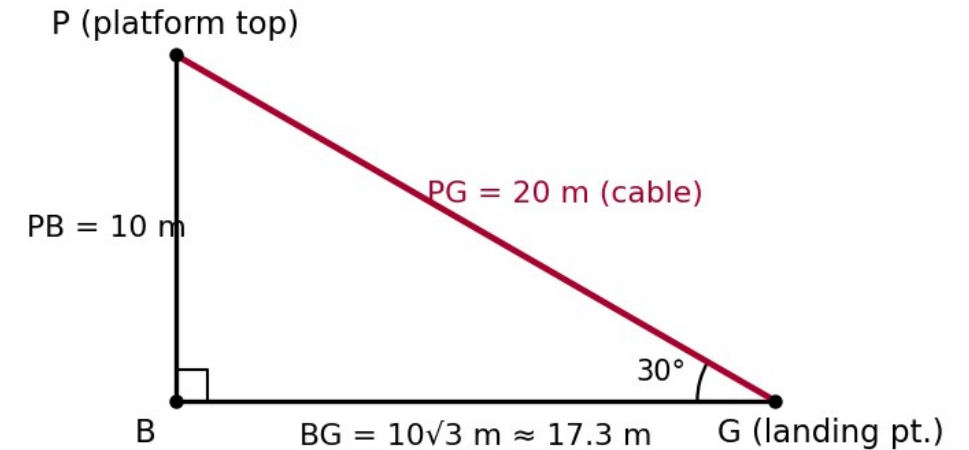
Shortcut: Since same-side interior angles always sum to 180, subtract the given angle from 180 in one step: $180 - 118 = 62$.

What It Is & Core Rules

- A right triangle is any triangle with exactly one 90-degree angle, marked by a small square in diagrams.
- The Pythagorean theorem links the two shorter sides to the longest side in every right triangle.
- Trigonometric ratios (sine, cosine, tangent) connect an angle to the ratio of two of the triangle's sides.
- The SAT tests this because right triangles model real heights, distances, ramps, and shadows.
- Two special right triangles let you find every side instantly, without ever squaring or using a calculator.
- The Pythagorean Theorem — $a^2 + b^2 = c^2$
- Pythagorean Triples — 3-4-5, 5-12-13, 8-15-17, 7-24-25
- 45-45-90 Triangle — legs: x, x hypotenuse: $x\sqrt{2}$
- 30-60-90 Triangle — sides: $x, x\sqrt{3}, 2x$ (opposite 30, 60, 90)
- SOH-CAH-TOA — $\sin\theta = \text{Opp}/\text{Hyp}$, $\cos\theta = \text{Adj}/\text{Hyp}$, $\tan\theta = \text{Opp}/\text{Adj}$
- Similar Right Triangles — AA, SSS, or SAS similarity
- Complementary Angles & the Pythagorean Identity — $\sin(90^\circ - \theta) = \cos\theta$. $\sin^2\theta + \cos^2\theta = 1$

Common Mistakes: Mixing up opposite and adjacent — relabel both sides fresh for whichever angle the problem names. ⚠ Assuming a triangle is right just because it looks that way — always confirm the marked right angle.

A zip-line cable in Salzburg runs from a launch platform down to a landing point on the ground, making a 30-degree angle with the ground at the landing point. Label the landing point G, the point on the ground directly below the platform B (right angle here), and the top of the platform P. The cable PG is the hypotenuse and is 20 meters long. Side PB (vertical, opposite the 30-degree angle at G) is the platform's height. Side BG (horizontal, opposite the 60-degree angle at P) is the horizontal distance a rider covers. Find BG, the horizontal distance covered.



Formula / Rule Used: 30-60-90 side ratios: x (opp. 30 deg), $x\sqrt{3}$ (opp. 60 deg), $2x$ (opp. 90 deg, hypotenuse)

Work through it, then advance for the full step-by-step solution.



Step 1: In a 30-60-90 triangle, the shortest side (opposite the 30-degree angle) is always half the hypotenuse — call it x .

Step 2: The hypotenuse (cable PG) is 20, so $x = 20 / 2 = 10$. This x is the platform's height, PB — not yet what's asked.

Step 3: The side opposite the 60-degree angle (BG, the horizontal distance) always equals $x \cdot \sqrt{3}$.

Step 4: Substitute $x = 10$: $BG = 10 \cdot \sqrt{3}$.

Step 5: As a decimal, $\sqrt{3}$ is about 1.732, so $10 \cdot 1.732$ is about 17.3.

Step 6: The horizontal distance covered is $10 \cdot \sqrt{3}$ meters, about 17.3 meters.

Answer: $10 \cdot \sqrt{3}$ meters (about 17.3 meters)

Tip: In a 30-60-90 word problem, first find the shortest side (half the hypotenuse) — every other side is built from it.

Trap: Stopping after finding the shortest side (10, the platform's height) and reporting that as the final answer, instead of continuing on to find the side opposite 60 degrees.



Tempting Wrong Answers — and Exactly What Went Wrong

10

Stopped after finding the 30-degree-opposite side (the platform height) without continuing to the 60-degree-opposite side that the question actually asks for.

40

Doubled the hypotenuse instead of halving it to find x , throwing off every side that follows.

$20\sqrt{3}$, about 34.6

Used the full hypotenuse (20) times $\sqrt{3}$ instead of the shortest side (10) times $\sqrt{3}$.

Shortcut: Once you know one side of a 30-60-90 triangle, every other side is a fixed multiple away — no need to set up sine or cosine at all.

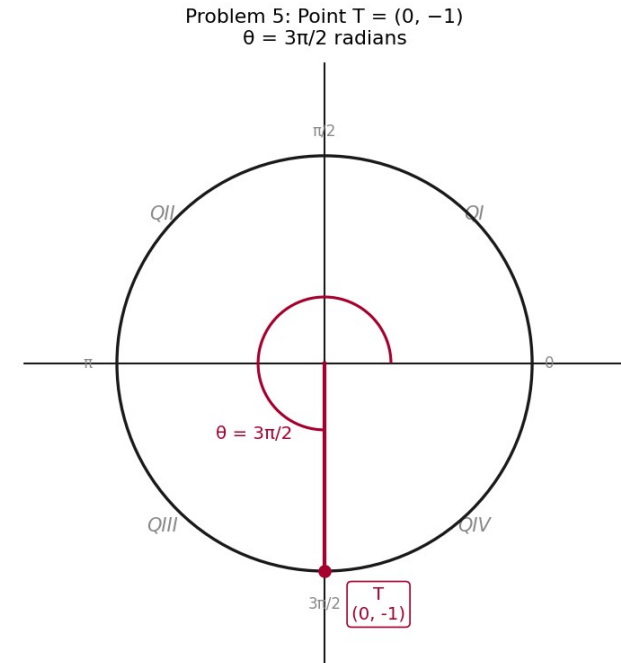
Desmos Tip: Type $10\sqrt{3}$ into the calculator to confirm the horizontal distance is about 17.3 meters.

What It Is & Core Rules

- A radian is just another way to measure an angle, like degrees, but based on the circle's own radius.
- The unit circle is a circle with radius 1, centered exactly at the origin (0, 0).
- Any point on the unit circle can be found using just the angle from the positive x-axis.
- The SAT tests this because it connects angles, coordinates, and the trig functions sine and cosine in one picture.
- You'll see radians in equations, and unit-circle points as coordinates you must identify or use.
- Degree-Radian Conversion — π radians = 180 degrees
- Common Angles Table — $30=\pi/6$, $45=\pi/4$, $60=\pi/3$, $90=\pi/2$, $180=\pi$, $270=3\pi/2$, $360=2\pi$
- The Unit Circle Itself — $x^2 + y^2 = 1$, centered at the origin, radius 1
- Coordinates from an Angle — point at angle $\theta = (\cos(\theta), \sin(\theta))$
- Key Reference Points — 0 deg=(1,0), 90 deg=(0,1), 180 deg=(-1,0), 270 deg=(0,-1)
- Quadrant Signs — QI (+,+), QII (-,+), QIII (-,-), QIV (+,-)
- Coterminal Angles — add or subtract 360 degrees (or 2π radians)

Common Mistakes: Forgetting to flip the conversion fraction, multiplying by $180/\pi$ when going from degrees to radians instead of $\pi/180$. ⚠ Mixing up which coordinate is cosine and which is sine, since x always goes with cosine and y always goes with sine.

Point T lies on the unit circle $x^2 + y^2 = 1$, centered at the origin O. The terminal side of angle theta, measured counterclockwise from the positive x-axis, meets the circle exactly at point T with coordinates (0, -1). If $0 \leq \theta < 2\pi$, what is the value of theta in radians?



Formula / Rule Used: Point at angle theta = (cos theta, sin theta); key reference points.

Work through it, then advance for the full step-by-step solution.

Step 1: The coordinates (0, -1) mean $\cos(\theta) = 0$ and $\sin(\theta) = -1$.

Step 2: $\cos(\theta) = 0$ only happens at the top or bottom of the circle, so θ is $\pi/2$ or $3\pi/2$.

Step 3: $\sin(\theta) = -1$ means the point is at the very bottom of the circle, not the top.

Step 4: The point straight down from the origin is reached at $\theta = 3\pi/2$ radians (270 degrees).

Step 5: Check: $\cos(3\pi/2) = 0$ and $\sin(3\pi/2) = -1$, which matches point T exactly.

Answer: $\theta = 3\pi/2$ radians

Tip: When you're given coordinates instead of an angle, first check which axis the point sits on, then use the key reference points to name the angle.

Trap: A common mistake is picking $\theta = \pi/2$ because $\cos(\theta) = 0$ there too, without checking that $\sin(\theta) = 1$ (not -1) at that angle.

Tempting Wrong Answers — and Exactly What Went Wrong

$\pi/2$

Matched $\cos(\theta) = 0$ correctly but ignored that $\sin(\pi/2) = 1$, not -1 , so the sign didn't check out.

π

Confused the bottom point $(0, -1)$ with the left point $(-1, 0)$, mixing up which axis the coordinates lie on.

270

Found the correct angle but reported it in degrees instead of the radians the question asked for.

Shortcut: Since $(0, -1)$ is one of the four key axis points, you can name θ straight from memory without setting up $\cos$ and $\sin$ equations at all.

Desmos Tip: Type $x^2 + y^2 = 1$ to draw the unit circle, then type the point $(0, -1)$ to see exactly where it lands, confirming it's the bottom of the circle at $(3\pi)/2$.

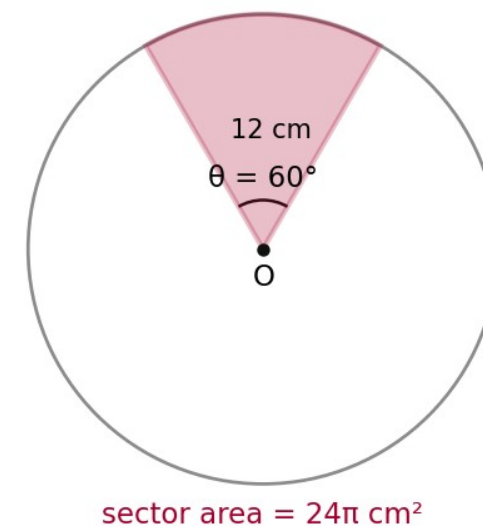
What It Is & Core Rules

- A circle is every point that sits the exact same distance (the radius) from one fixed center point.
- The SAT tests circles because a small set of formulas can answer many different-looking questions.
- Arc length and sector area are always just a fraction of the whole circle's circumference or area.
- Circles also show up as algebra: an equation like $(x - h)^2 + (y - k)^2 = r^2$ describes the same shape.
- A few angle theorems (tangent lines, inscribed angles) connect straight-line angles to the circle itself.
- Circumference and Area — $C = 2\pi r = \pi d$. $A = \pi r^2$
- Arc Length — Arc length = $(\theta/360^\circ) \times 2\pi r$
- Sector Area — Sector area = $(\theta/360^\circ) \times \pi r^2$
- Equation of a Circle — $(x - h)^2 + (y - k)^2 = r^2$
- Completing the Square for a Circle
- Tangent Lines — radius $\perp$ tangent line
- Inscribed Angle Theorem — inscribed angle = $(1/2) \times$ intercepted arc

Common Mistakes: Using a chord length instead of the actual central angle inside the $\theta/360^\circ$ fraction. ⚠ Forgetting the sign flip in the circle equation, writing $(x + h)$ instead of $(x - h)$ for a positive h .

A circular clock face in a train station in Brussels has a radius of 12 centimeters. A decorative sector cut from the clock face - like a pie slice bounded by two radii and an arc - has an area of 24π square centimeters. What is the measure of the sector's central angle, in degrees?

Clock face: $r = 12$ cm, sector area = 24π



Formula / Rule Used: Sector area = $(\theta/360)^\circ \times \pi r^2$, solved backward for θ .

Work through it, then advance for the full step-by-step solution.



Step 1: Start from the sector area formula: $\text{sector area} = (\theta/360) \cdot \pi r^2$.

Step 2: Find the full circle's area first: $\pi r^2 = \pi(12)^2 = 144\pi$ square centimeters.

Step 3: Substitute the known sector area into the formula: $24\pi = (\theta/360) \times 144\pi$.

Step 4: Divide both sides by π to cancel it: $24 = (\theta/360) \times 144$.

Step 5: Divide both sides by 144 to isolate the fraction: $\theta/360 = 24/144$.

Step 6: Simplify the fraction: $24/144 = 1/6$.

Step 7: Multiply both sides by 360 to solve for θ : $\theta = 360 \times (1/6) = 60$.

Answer: 60°

Tip: When you're given the sector's area instead of the angle, set up the same $(\theta/360) \cdot \pi r^2$ equation and solve backward for θ .

Trap: It's tempting to divide the sector area by the radius (12) instead of by the full circle's area (144π), which skips squaring the radius entirely.



Tempting Wrong Answers — and Exactly What Went Wrong

720°

Divided 24π by 12 (the radius) instead of by $\pi r^2 = 144\pi$, then never checked that a central angle must be under 360° .

180°

Set the fraction up as $\frac{24}{48}$ (using $4r$ instead of $r^2 = 144$) instead of the correct $\frac{24}{144}$.

48°

Divided 24 by 144 correctly but then multiplied by 288 instead of 360, using a miscopied denominator.

Shortcut: Notice 24π is exactly $\frac{1}{6}$ of the full area 144π at a glance, so $\theta = 360/6 = 60^\circ$ without writing out the full equation.

Desmos Tip: Type $24/144 * 360$ into Desmos as a quick arithmetic check to confirm $\theta = 60$.

What It Is & Core Rules

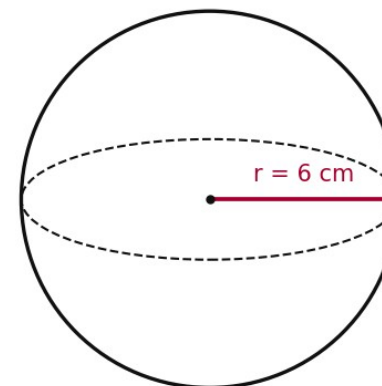
- Volume measures how much 3D space is inside a solid, like how much water a container could hold.
- Surface area measures the total area of every outside face of a solid, like wrapping paper needed to cover it.
- The SAT tests these because a handful of formulas can answer almost any 3D-shape question it asks.
- Similar solids are the same shape scaled up or down, so a single scale factor connects their lengths, areas, and volumes.
- Most volume formulas reduce to 'area of the base times height,' sometimes with an extra fraction out front.
- Prisms and Cylinders — $V = (\text{area of base}) \times \text{height}$
- Pyramids and Cones — $V = (1/3) \times (\text{area of base}) \times \text{height}$
- Spheres — $V = (4/3)\pi r^3$. Surface area = $4\pi r^2$
- Surface Area of Common Solids — Cube: $S = 6s^2$. Rect. prism: $S = 2(lw + lh + wh)$. Cylinder: $S = 2\pi rh + 2\pi r^2$
- Composite Solids
- Polygon Interior Angle Sum — $(n - 2) \times 180^\circ$
- Similarity Ratios — length ratio = λ . area ratio = λ^2 . volume ratio = λ^3

Common Mistakes: Forgetting the 1/3 factor for pyramids and cones, and using the prism/cylinder formula instead. ⚠ Plugging the diameter directly into a formula that actually needs the radius (half the diameter).

A spherical stress ball has a surface area of 144π square centimeters. What is the volume of the ball, in cubic centimeters, in terms of π ?

Sphere — Stress Ball

$$\text{Surface area } S = 4\pi r^2 = 144\pi \text{ cm}^2$$



Formula / Rule Used: Sphere surface area ($S = 4\pi r^2$), then sphere volume ($V = (4/3)\pi r^3$).

Work through it, then advance for the full step-by-step solution.



Step 1: This problem works backward: set the surface area formula equal to the given value: $4\pi r^2 = 144\pi$.

Step 2: Divide both sides by 4π to isolate r^2 : $r^2 = 144/4 = 36$.

Step 3: Take the square root of both sides to find r : $r = 6$.

Step 4: Now switch to the volume formula and plug in $r = 6$: $V = (4/3)\pi(6)^3$.

Step 5: Cube the radius: $6^3 = 216$.

Step 6: Multiply: $(4/3) \times 216 = 288$.

Step 7: The volume is 288π cubic centimeters.

Answer: 288π cubic centimeters

Tip: When a sphere problem gives surface area but asks for volume, always solve for the radius first, then switch formulas.

Trap: After finding $r^2 = 36$ from the surface area equation, it's tempting to plug 36 straight into the volume formula instead of $r = 6$.

Tempting Wrong Answers — and Exactly What Went Wrong

48π

Used $r^2 = 36$ directly in the volume formula instead of finding $r = 6$ and cubing it, computing $(4/3)\pi(36)$.

864π

Forgot to divide by 3, computing $4\pi(216)$ instead of $(4/3)\pi(216)$.

36π

Treated the radius found (6) as if it were a diameter, halving it to 3 before cubing.

Shortcut: Since $144/4 = 36$ is a perfect square, $r = 6$ comes out cleanly without a calculator.

Desmos Tip: Type $\sqrt{(144/4)}$ in Desmos to confirm $r = 6$, then type $(4/3) \cdot 3.14159 \cdot 6^3$ to confirm the volume is about 904.8, matching 288π .